THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 3 Solution

1. Suppose that an airplane engine will fail, when in flight, with probability $1-p$ independently from engine to engine. Assume that the airplane will make a successful flight if at least 50 percent of its engines remain operative. For what values of p are a four-engine plane preferable to a two-engine plane?

   Solution:

   Let X_n be the number of working engines for an n-engine plane. Using a combinatorial argument, there are $\binom{n}{k}$ ways to choose k engines to work. For each way, since we require precisely k working engines and $n-k$ non-working engine, it happens with probability $p^k(1-p)^{n-k}$. Therefore, by the additivity of probability,

   $$\mathbb{P}(X_n = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n. \tag{1}$$

   We are going to find p such that a four-engine plane is preferable to a two-engine plane, i.e. $\mathbb{P}(X_4 \ge 2) \ge \mathbb{P}(X_2 \ge 1)$. By the above discussion, we have

   $$\begin{aligned}\mathbb{P}(X_4 \ge 2) &= \binom{4}{2} p^2(1-p)^2 + \binom{4}{3} p^3(1-p) + \binom{4}{4} p^4(1-p)^0 \\ &= 6p^2(1-p)^2 + 4p^3(1-p) + p^4.\end{aligned}$$

   Similarly,

   $$\begin{aligned}\mathbb{P}(X_2 \ge 1) &= \binom{2}{1} p^1(1-p)^1 + \binom{2}{2} p^2(1-p)^0 \\ &= 2p(1-p) + p^2.\end{aligned}$$

   Hence the four-engine plane is safer if

   $$6p^2(1-p)^2 + 4p^3(1-p) + p^4 \ge 2p(1-p) + p^2,$$

   which simplifies to

   $$p(1-p)^2(3p-2) \ge 0.$$

   Recall that we always have the restriction $0 \le p \le 1$. Therefore the four-engine plane is preferable to the two-engine plane if $\frac{2}{3} \le p \le 1$.

   Remark: We call any random variable satisfying (1) a *binomial* random variable.

2. A coin having probability p of coming up heads is successively flipped until the r-th head appears. Suppose that the results of flips are mutually independent. Argue that X, the number of flips required, will be n with probability

   $$\mathbb{P}(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \quad n \ge r.$$

   This is known as the *negative binomial distribution* with parameters p and r.

   Solution:

   Note that the event $\{X = n\}$ is characterized by the unions of all outcomes satisfying

(a) The first $n-1$ flips contain exactly $r-1$ heads and $n-r$ tails.

(b) The n-th flip is a head.

Since we require r heads and $n-r$ tails in total, each pattern happens equally likely with probability $p^r(1-p)^{n-r}$. A combinatorial argument shows that there are $\binom{n-1}{r-1}$ distinct ways to choose the $r-1$ heads among the first $n-1$ flips. Thus we conclude that $\mathbb{P}(X=n) = \binom{n-1}{r-1}p^r(1-p)^{n-r}$.

3. Suppose that independent trials, each of which results in any of m possible outcomes with respective probabilities $p_1, \ldots, p_m$, $\sum_{i=1}^m p_i = 1$, are continually performed. Let X denote the number of trials needed until each outcome has occurred at least once.

(a) If $m=2$, show that for any $n \geq 1$, we have

$$\mathbb{P}(X > n) = (1-p_1)^n + (1-p_2)^n.$$

(b) Find $\mathbb{P}(X > n)$ for general $m \geq 2$.

(c) Find $\mathbb{P}(X = n)$.

Hint: You may use the result in Homework 1 Question 1.

Solution:

Let A_i denote the event that the outcome i has not yet occurred after the first n trials, $i = 1, \ldots, m$.

(a) We have

$$\begin{aligned}\mathbb{P}(X > n) &= \mathbb{P}\left(A_1 \cup A_2\right) \\ &= \mathbb{P}(A_1) + \mathbb{P}(A_2) - \mathbb{P}(A_1 \cap A_2).\end{aligned}$$

Note that for each $n \geq 1$, by the independence of each trial,

$$\mathbb{P}(A_i) = (1-p_i)^n, \quad i = 1, 2,$$

For each trial, neither outcome 1 nor outcome 2 occurs with probability $1 - p_1 - p_2$. Therefore, the probability that neither outcome 1 nor outcome 2 occurs in the first n trials is

$$\mathbb{P}(A_1 \cap A_2) = (1-p_1-p_2)^n = 0.$$

Therefore,

$$\mathbb{P}(X > n) = (1-p_1)^n + (1-p_2)^n.$$

(b) Generalizing the above argument, we have

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = (1 - (p_{i_1} + p_{i_2} + \cdots + p_{i_k}))^n$$

for all $i_1, i_2, \ldots, i_k$ such that $1 \leq i_1 < i_2 < \cdots < i_k \leq m$. Leveraging the principle of

inclusion-exclusion (i.e. the result in Homework 1 Q1), we have

$$\begin{aligned}
\mathbb{P}(X > n) &= \sum_{i_1=1}^{m} \mathbb{P}(A_{i_1}) - \sum_{1\le i_1<i_2\le m} \mathbb{P}(A_{i_1} \cap A_{i_2}) + \sum_{1\le i_1<i_2<i_3\le m} \mathbb{P}(A_{i_1} \cap A_{i_2} \cap A_{i_3}) \\
&\quad - \cdots + (-1)^m \sum_{1\le i_1<i_2<\cdots<i_{m-1}\le m} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_{m-1}}) \\
&\quad + (-1)^{m+1} \mathbb{P}(A_1 \cap \cdots \cap A_m) \\
&= \sum_{i_1=1}^{m} (1-p_{i_1})^n - \sum_{1\le i_1<i_2\le m} (1-p_{i_1}-p_{i_2})^n + \sum_{1\le i_1<i_2<i_3\le m} (1-p_{i_1}-p_{i_2}-p_{i_3})^n \\
&\quad - \cdots + (-1)^m \sum_{1\le i_1<i_2<\cdots<i_{m-1}\le m} (1-p_{i_1}-\cdots-p_{i_{m-1}}) + 0 \\
&= \sum_{k=1}^{m-1} \sum_{1\le i_1<i_2<\cdots<i_k\le m} (-1)^{k+1} \left(1 - \sum_{\ell=1}^{k} p_{i_\ell}\right)^n .
\end{aligned}$$

(c) Note that

$$\mathbb{P}(X = n) = \mathbb{P}(X > n-1) - \mathbb{P}(X > n).$$

Using the algebraic identity $(1-a)^{n-1} - (1-a)^n = a(1-a)^{n-1}$ and taking $a = \sum_{\ell=1}^{k} p_{i_\ell}$, the result simplifies to

$$\mathbb{P}(X = n) = \sum_{k=1}^{m-1} \sum_{1\le i_1<i_2<\cdots<i_k\le m} (-1)^{k+1} \left(\sum_{\ell=1}^{k} p_{i_\ell}\right) \left(1 - \sum_{\ell=1}^{k} p_{i_\ell}\right)^{n-1} .$$

4. (a) Prove the memoryless property of geometric random variables. i.e., if $X \sim Geom(p)$, then

$$P(X = n+k \mid X > k) = P(X = n), \quad \forall n, k \ge 0.$$

(b) Prove the memoryless property of exponential random variables. i.e., if $X \sim Exp(\lambda)$, then

$$P(X > n+k \mid X > k) = P(X > n), \quad \forall n, k \ge 0.$$

Solution:

(a) The equality holds trivially when $n = 0$. For $n \ge 1$, note that $X = n + k$ and $X > k$ if and only if $X = n + k$. Thus,

$$\begin{aligned}
P(X = n+k \mid X > k) &= \frac{P(X = n+k \cap X > k)}{P(X > k)} = \frac{P(X = n+k)}{1 - \sum_{i=1}^{k} P(X = i)} \\
&= \frac{(1-p)^{n+k-1} p}{(1-p)^k} \\
&= (1-p)^{n-1} p \\
&= P(X = n).
\end{aligned}$$

(b) The equality holds trivially when $n = 0$. For $n > 0$, note that $X = n + k$ and $X > k$ if and only if $X = n + k$. Recall that the cumulative distribution function of an exponential random variable is given by $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. Therefore, we have Thus,

$$\begin{aligned} P(X > n + k \mid X > k) &= \frac{P(X > n + k \cap X > k)}{P(X > k)} = \frac{P(X > n + k)}{P(X > k)} \\ &= \frac{1 - (1 - e^{-\lambda(n+k)})}{1 - (1 - e^{-\lambda k})} = e^{-\lambda n} \\ &= 1 - (1 - e^{-\lambda n}) \\ &= P(X > n). \end{aligned}$$